Robotics-for-Cacao Barter

Follow-up action item · TrueSight DAO / Agroverse · 2026-08-06

Context. Three field priorities: harvesting cacao, cutting off witches' broom (*vassoura-de-bruxa*, *Moniliophthora perniciosa*), and cacao roasting flavor-profile control. A robotics design expert partner is on board. Working-prototype reference: **Frasky**, the IIT vineyard robot (Reuters video, https://youtu.be/hg8qYrjyYCU). This follow-up turns that introduction into a concrete pilot.

1 · The problem on the farm

- **Witches' broom** is the single highest-leverage canopy disease in Bahia cacao: infected brooms must be pruned out **and removed from the plantation** in dry season, or spores reinfect — cutting yields by 30–50% on affected trees.
- **Labor shortage**: pruning and selective pod harvest are manual, seasonal, and hard to staff.
- **Cash constraint**: smallholders have little cash but hold **cacao** — the natural payment asset.

2 · Working prototype reference — Frasky (IIT)

Video (Reuters): https://youtu.be/hg8qYrjyYCU — "A hands-on robot tends grapes in Italy's vineyards"

- Built by the **Italian Institute of Technology + Bergamo-region agricultural stakeholders** (roboticist Dr. Manuel Catalano).
- Autonomously **monitors grape clusters** (camera + digital map of every cluster's location), **manipulates plants** (robotic arm), and **applies targeted spray treatments**.
- Designed for precision agriculture, **labor-shortage relief**, and sustainability — field-tested in vineyards (Reuters, Nov 2025).

Proof point: a permanent tree/vine crop's canopy work can be robotized. Cacao is the same category — this is the working prototype we point partners and farmers to.

Frasky capability (grape)	Cacao adaptation	Feasibility
Cluster monitoring via camera	Broom / pod detection via vision	■ tractable
Manipulate plants (robotic arm)	Broom pruning with cutter arm	■ tractable
Targeted spray treatments	Targeted biocontrol on brooms	■ tractable
—	Selective pod *harvest* (varied heights: trunk + branches)	■■ harder R&D

3 · Second application — roast control as a reinforcement learning problem

The gap. Cacao roasting — the step that defines the flavor profile — is a **sequential decision process with delayed reward**: the operator acts on the roast curve (heat, airflow, drum speed, time), and only *after* the batch do you learn whether the flavor profile landed. Today that is skilled-craft, tuned batch-by-batch by experienced tasters — **not scalable** beyond artisan volumes, and hard to reproduce consistently.

The reframe — this is a reinforcement learning problem.

RL element	In the cacao roaster
State (observation)	Bean moisture, bean temperature / rate-of-rise, roast color, particle size distribution, volatile markers
Action	Heat input, airflow, drum speed, batch timing — the roast-curve decisions
Reward	Flavor-profile match (sensor + taste panel), batch-to-batch consistency, energy efficiency
Policy	A learned roast curve that reproduces a target flavor profile, batch after batch

The agent **learns the roast policy** over batches — each batch is a training episode, the reward comes at the end. That is exactly what makes it RL rather than a static recipe: the controller adapts to bean lot, moisture, and machine state to hit the flavor target.

The observation layer — particle / roast-state detection. The RL agent is only as good as what it can sense:

Detection approach	What it measures	Status
Machine vision (camera)	Bean color, surface browning, particle size distribution	■ proven in food lines
Hyperspectral / SWIR imaging	Moisture, roast degree, bean defects (slaty, moldy, under/over-fermented)	■ proven, conveyor-scalable
Portable NIR spectroscopy	Chemical markers of roast degree, in-line	■ proven, low-cost
Electronic nose + ANN	Volatile profile → predicted roast degree (94.4% accuracy reported)	■ proven

Why now / why us. Precedent exists (IMA Group's "AI Learning to Roast", model-based roast optimization) but is sold as expensive industrial software. A robotics + ML partner can build a **low-cost RL roast controller + particle detector** tuned to the beans and flavor profiles the Agroverse network actually runs — turning roast flavor control from craft into a **scalable, self-improving process**.

This is the **second application** for the same robotics partner: field robots (pruning/harvest) + processing-side RL roast control (particle detection as the observation layer).

4 · The proposal (barter model)

Offer robotics services to Bahia/Pará farmers **in exchange for cacao as payment**:

Role	Who
Robot design & build	Robotics design expert (Gianluca)
Farmer network & deployment	Agroverse — Bahia operations (Ilhéus)
Market for cacao received	TrueSight DAO / Agroverse (US import live)

Honest MVP scoping. Broom *detection* + *pruning* (or monitoring + targeted treatment, à la Frasky) is tractable now. *Selective cacao pod harvesting* is harder R&D — set expectations so the pilot stays deliverable. Roast RL control is a parallel, lower-field-risk track.

5 · Pilot plan (1 season, measured)

1. **Scoping call with the robotics partner** — share the **Frasky video** (https://youtu.be/hg8qYrjyYCU) as the reference prototype; define MVP build, season timing, and cost basis; evaluate the roast-RL track (state sensors, reward signal, data capture per batch).
2. **Farmer pitch (PT-BR)** — value prop: labor shortage, broom yield loss, **no cash outlay**, paid in cacao at fair market BRL.
3. **2–3 test farms** in the Bahia (Ilhéus) network.
4. **Barter accounting** — value cacao received at market BRL/USD; record as INVENTORY MOVEMENT + purchase into the Main Ledger (see SUPPLY_CHAIN_AND_FREIGHTING.md).
5. **Measure, then scale** — % broom removed, harvest kg, labor-hours saved; roast batch-to-batch consistency. No scaling on unmeasured evidence.

6 · Next steps (owners)

- **Approve outreach copy** and introduce the robotics partner (WhatsApp).
- **Draft scoping-call question list** (incl. Frasky feasibility + roast-RL questions: observation sensors, reward definition, data per batch) and the PT-BR farmer pitch.
- **Robotics partner** — feasibility call; MVP definition.
- **Agroverse Brazil operations** — identify 2–3 test farms; coordinate season logistics.

7 · Tie-back to mission

Every broom pruned, every robot-assisted harvest, and every batch of cacao roasted to a reproducible flavor profile keeps more trees producing and more **Amazon rainforest standing** — the DAO's 10,000-hectare regeneration goal is served by making cacao farming viable without clearing new land.

Filed in agentic_ai_context/OPEN_FOLLOWUPS.md (PR #733) · nothing sent to any party yet — awaiting governor approval of outreach copy.